

**2022**

**YEAR 11 YEARLY EXAMINATION**

# **Mathematics Standard**

**Examiner: Sami El Hosri**

## **General Instructions**

- Reading Time – 10 minutes
- Working Time – 2 hours
- Write using black pen
- Calculators approved by NESAs may be used
- A separate reference sheet is provided at the back of this paper.
- All necessary working should be shown in Questions 16 – 35

## **Total marks – 75**

### **Section I: Multiple Choice**

#### **Questions 1 – 15 15 marks**

- Attempt all questions
- Allow about 25 minutes for this section

### **Section II: Extended Response**

#### **Questions 16 – 35 60 marks**

- Attempt all questions
- Allow about 1 hour 35 minutes for this section

## Section I

### Questions 1 – 15 (15 marks)

Read each question and choose an answer A, B, C or D.

Allow about 25 minutes for this section.

Use the multiple-choice answer sheet for Questions 1-15.

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- 1 A sphere has a radius of 5 cm.

What is the volume of this sphere correct to 2 decimal places?

A)  $261.80 \text{ cm}^3$

B)  $523.60 \text{ cm}^3$

C)  $314.16 \text{ cm}^3$

D)  $534.59 \text{ cm}^3$

- 2 Darren is a restaurant manager.

The table below shows some information about the hours he worked last weekend and his pay per hour.

| Day      | Hours Worked | Pay per hour | Payment     |
|----------|--------------|--------------|-------------|
| Saturday |              | \$45         |             |
| Sunday   | 9            | \$55         |             |
|          |              |              | Total \$855 |

How many hours did Darren work on Saturday?

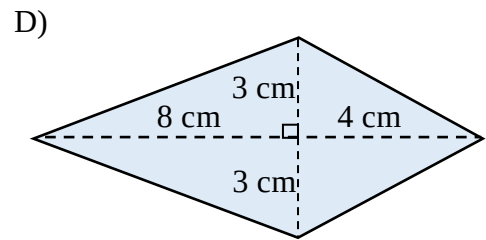
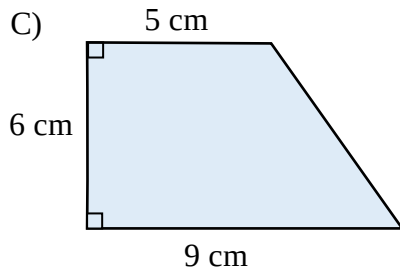
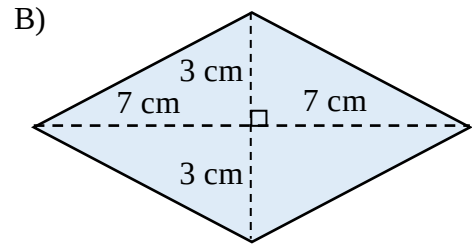
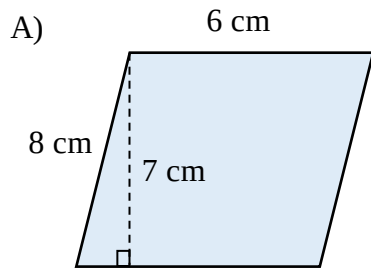
A) 6

B) 7

C) 8

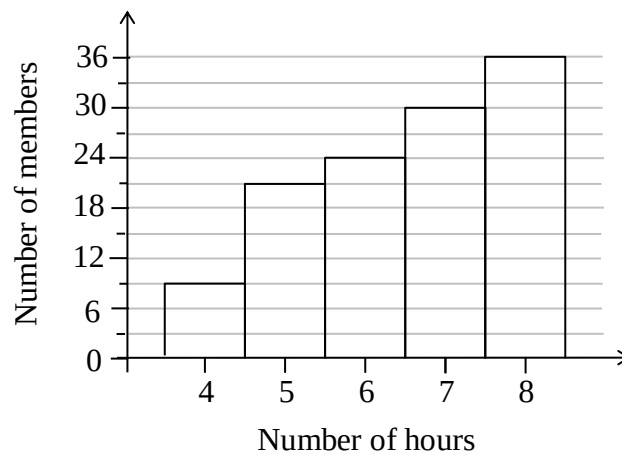
D) 9

3 Which of the following shapes has the smallest area?



4 Members of a swimming club were surveyed about the number of hours they spent training last Sunday.

The results are shown in this cumulative frequency histogram.



How many members said they have spent 5 hours training last Sunday?

- A) 12
- B) 14
- C) 20
- D) 21

- 5 City A is located at (12°S, 82°E) and City B is located at (48°N, 23° W).

What is the day and time in City A when it is 10 pm Monday in City B?

You can use  $15^\circ = 1$  hour time difference and ignore daylight saving.

- A) 6 pm Monday
- B) 3 pm Monday
- C) 2 am Tuesday
- D) 5 am Tuesday

- 6 Amanda went on holidays for 10 days. Her average spending during the first 4 days is \$125 per day.

Her overall average spending is \$140 per day.

What is her average spending during the last six days?

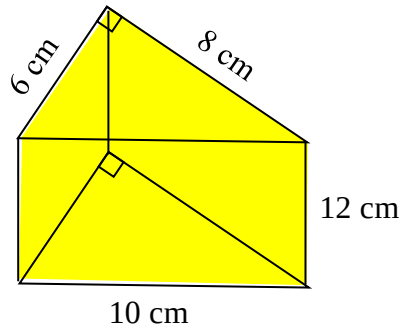
- A) \$125
- B) \$150
- C) \$132.50
- D) \$90

- 7 A box contains 9 whiteboard markers. 5 of these markers are black and 4 are blue. Jacob is to select 2 markers from the box at random.

What is the probability that Jacob selects 2 blue markers?

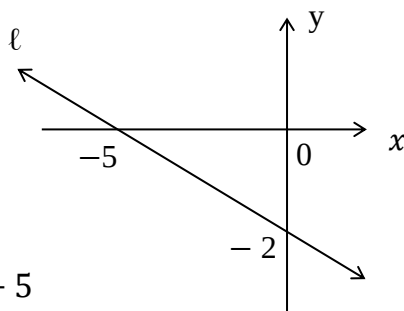
- A)  $\frac{4}{9} \times \frac{3}{8}$
- B)  $\frac{4}{9} + \frac{3}{9}$
- C)  $\frac{4}{9} + \frac{3}{8}$
- D)  $\frac{4}{9} \times \frac{3}{9}$

- 8 Charlie has a cheese block which is in the shape of a triangular prism. The dimensions of the cheese block are as shown in the diagram.



- What is the surface area of this cheese block?
- A)  $144 \text{ cm}^2$   
B)  $216 \text{ cm}^2$   
C)  $336 \text{ cm}^2$   
D)  $384 \text{ cm}^2$
- 9 Jenny says the length of her pool is 400 cm, to the nearest centimetre. What is the percentage error in her measurement?
- A)  $\pm 0.0125\%$   
B)  $\pm 0.25 \%$   
C)  $\pm 0.0025\%$   
D)  $\pm 0.125\%$
- 10 Lachlan has 4 000 mangoes. He gives his brother 200 mangoes and each week he plans to sell 250 mangoes. Which of the following equations shows the number of mangoes  $N$  that Lachlan still has after  $p$  weeks?
- A)  $N = 3\,750 - 200p$   
B)  $N = 3\,800 - 250p$   
C)  $N = 4\,200 - 250p$   
D)  $N = 3\,800 + 250p$

- 11 The graph shows a line ( $\ell$ ).  
Which of the following is the equation of line ( $\ell$ )?



- A)  $y = -\frac{5}{2}x - 5$
- B)  $y = -\frac{5}{2}x - 2$
- C)  $y = -\frac{2}{5}x - 2$
- D)  $y = -\frac{2}{5}x - 5$
- 12 A sick person was given 0.8 L of fluid over 8 hours.  
He was given the fluid at a rate of 9 drops every 4 seconds.  
What is the rate of drops per mL at which the fluid was given to the person?
- A) 44.44 drops/mL
- B) 1.35 drops/mL
- C) 81 drops/mL
- D) 85 drops/mL
- 13 Kylie won a prize in a lottery. She invested the money she won in her prize at a simple interest rate of 4% per annum for 5 years.  
After 5 years, the total value of Kylie's investment was \$64 800.  
How much money did Kylie win in the lottery prize?
- A) \$ 54 000
- B) \$10 800
- C) \$51 840
- D) \$12 960

- 14** Amy has two bags A and B. Each of them contains  $x$  lollies. The lollies in each bag are either strawberry lollies or honey lollies and none of the bags has any other flavour lollies.

The ratio of honey lollies to strawberry lollies in bag A and bag B are 4 : 5 and 2 : 1 respectively.

Bag B contains 6 more honey lollies than bag A.

What is the value of  $x$ ?

- A) 54
- B) 45
- C) 36
- D) 27

- 15** The dosage of medication for children aged 1 to 2, based on an adult dosage, can be calculated using Fried's formula shown below.

$$\text{Child dosage} = \frac{\text{age of child (in months)} \times \text{adult dosage}}{150}$$

John and his cousin Tania are sick and they are to take a medicine that has an adult dosage of 30 mL.

John's dosage is 2 mL and Tania's dosage is 3 mL.

By how many months is Tania older than John?

- A) 5 months
- B) 8 months
- C) 10 months
- D) 15 months

# **Year 11 MATHEMATICS STANDARD 2**

## **Section II Answer Booklet**

60 marks

Attempt Questions 16 – 35

Allow about 1 hour and 35 minutes for this section.

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### **INSTRUCTIONS**

- Answer the questions in the spaces provided. Sufficient space is provided for typical responses.
  - Your responses should include relevant mathematical reasoning and/or calculations.
  - Extra writing space is provided at the back of this booklet.
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**Question 16 (2 marks)**

A boat was purchased 3 years ago. Its value depreciates by \$950 each year.

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The value of this boat now is \$7 200.

What was the purchase price?

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**Question 17 (2 marks)**

Paul surveyed seven people. He asked them how much they paid for gas in their last bill.

Here are the results: \$350, \$270, \$332, \$460, \$342, \$280, \$282.

a) Calculate the mean of this set of data.

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b) What is the standard deviation of this set of data, correct to two decimal places?

1

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**Question 18 (2 marks)**

A plumber charges a call-out fee of \$50 and then \$80 per hour while working.

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If the job takes more than three hours, he discounts 10% of the total bill.

How much will this plumber charge for a 4 hour job ?

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**Question 19 (2 marks)**

Draw a possible graph for a negatively skewed distribution curve showing the possible location of the median and the mode.

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**QUESTION 20 (2 marks)**

Sara's car uses fuel at a rate of 9.2 L/100 km.

Rachel's car uses fuel at a rate of 7.6 L/100 km.

Last Sunday, Sara and Rachel each drove their cars for 120 km.

If the fuel cost is \$1.75 per litre, how much more did Sara pay than Rachel for fuel?

2

[illegible]

**Question 21 (2 marks)**

City P and City Q have Coordinated Universal Time of  $-3.00$  and  $+4.00$  respectively.

**2**

Pauline flew from City Q on a Monday at 1:20 am to go to City P.

The flight took 10 hours.

On what day and at what time did Pauline's plane arrive at City P?

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**Question 22 (3 marks)**

Kaydon bought a car with a cash price of \$12 000, on terms.

**3**

He agreed to pay 10% deposit and to pay the remaining amount plus interest over 20 months, in equal instalments.

If simple interest at the rate of 5% p.a. was charged on the remaining amount, with five months interest free, how much did Kaydon pay each month?

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**Question 23 (2 marks)**

Darren's fortnightly income is \$2 550. He is considering his fortnightly budget for the next 12 weeks to determine his total savings.

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His average expenses per fortnight over a period of 12 weeks is \$1580.

How much will Darren save in the next 12 weeks?

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**Question 24 (2 marks)**

Jane's yearly salary is \$90 000. She decides to take her four weeks annual holiday.

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Find her holiday pay, which consists of her normal pay for the four weeks plus 17.5% holiday loading. (Use 1 year = 52 weeks)

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### Question 25 (4 marks)

A set of discrete data has a lower quartile 12 and upper quartile 18. Its mean is 2 more than twice the interquartile range.

- (a) What is the mean of the data?

2

[illegible]

- (b) If this set of data has no outliers, what could be its maximum range? Justify your answer using suitable calculations.

2

[illegible]

**QUESTION 26 (2 marks)**

The profit  $P$  of a chicken shop varies directly with the number  $N$  of chickens sold.  
The profit is \$238 when 28 chickens are sold.

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What is the profit when 85 chickens are sold?

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**Question 27 (2 marks)**

A patient received 170 mL of a medicine in 1 hour and 25 minutes through  
an intravenous drip at a constant rate.

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How long in hours and minutes will it take for the patient to receive a dose of 238 mL?

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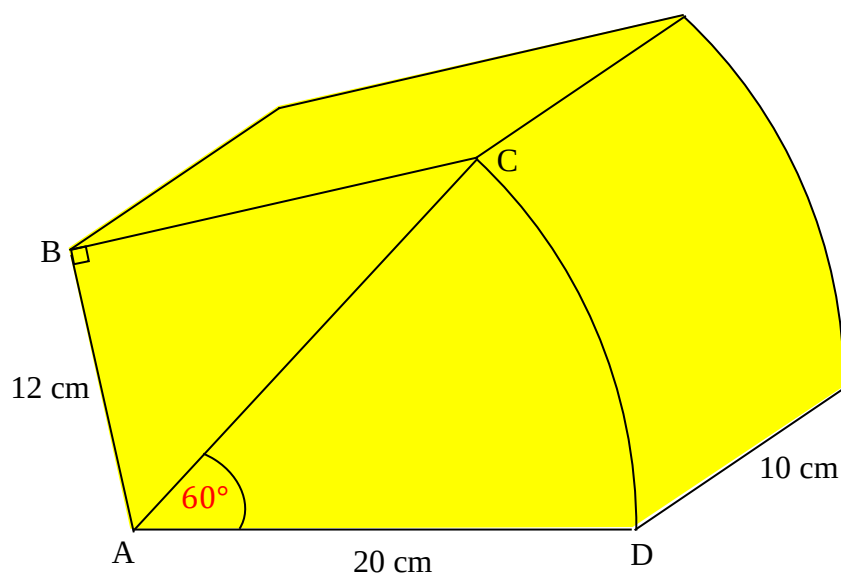
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**Question 28 (5 marks)**

The cross section of the prism shown in the diagram consists of the right-angled triangle ABC and the sector ACD.



(a) Find the area of the sector correct to two decimal places.

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(b) Hence, find the volume of the prism correct to one decimal place.

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**Question 29 (5 marks)**

Last financial year Daniel's taxable income was \$98 325, and his allowable deductions were 5% of his gross income.

**1**

a) How much was Daniel's gross income?

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b) Use the tax table below to calculate Daniel's tax on his taxable income.

**2**

| Taxable income       | Tax on this income                             |
|----------------------|------------------------------------------------|
| \$45,001 – \$120,000 | \$5 092 plus 32.5 c for each \$1 over \$45 000 |

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c) Calculate Daniel's Medicare levy which is 2% of his taxable income.

**1**

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d) After calculating his tax and Medicare levy, Daniel's was told he still needs to pay the tax office \$725 for his last financial year.

**1**

How much tax did he pay through his PAYG, throughout the last financial year?

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### Question 30 (7 marks)

Maria wants to hire a large boat for her wedding party. The boat company charges \$2500 for a booking fee to hire the boat and then adds an additional \$75 per guest for drink and food.

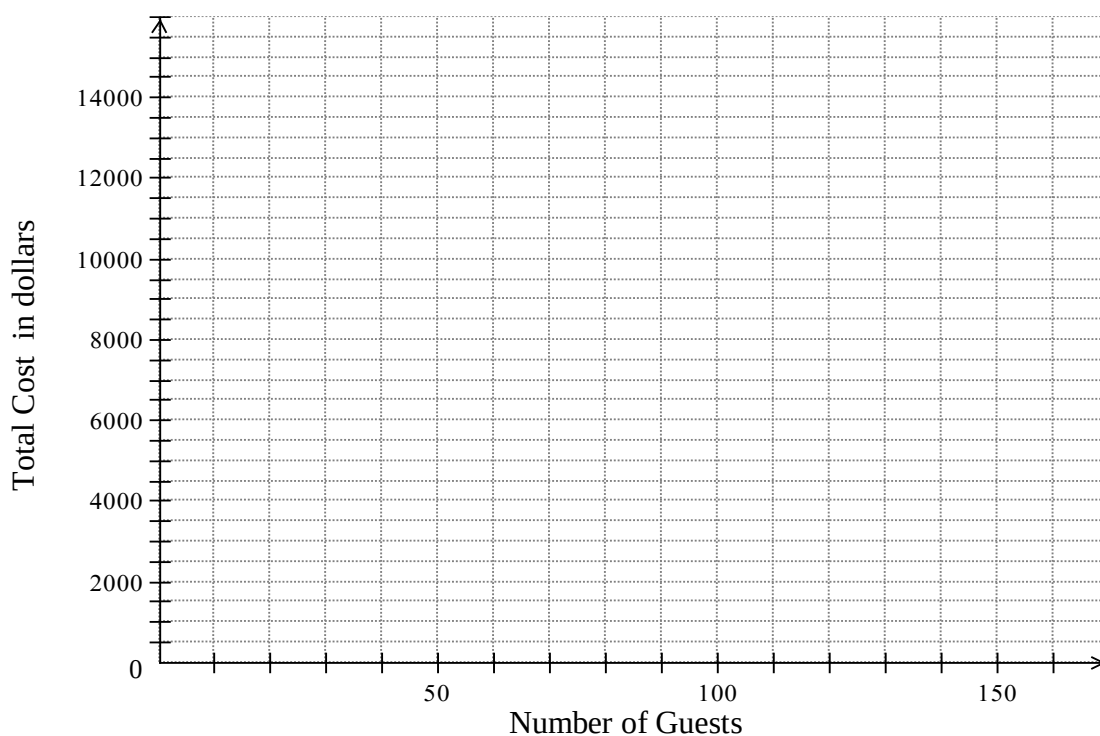
a) Use this information to complete the table below.

2

|                       |       |    |    |     |        |     |
|-----------------------|-------|----|----|-----|--------|-----|
| Number of Guests      | 0     | 40 | 80 | 100 | 120    | 140 |
| Total cost In dollars | 2 500 |    |    |     | 11 500 |     |

b) Plot the ordered pairs from the table above to graph the line representing the cost.

2



c) Find the equation which expresses the total cost  $C$  in terms of the number of guests  $N$ .

2

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d) Calculate the total cost to cater for a wedding party with 200 guests.

1

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### Question 31 (2 marks)

A can of corned beef contains 61.9 kilocalories due to the fat it contains.

2

What is the value of x the number of grams of fat in the can?

Use the following conversions. 1 gram of fat contains 37 kilojoules ( kJ) and 1 kilocalorie = 4.184 kilojoules.

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### Question 32 (3 marks)

Patrick has a mass of 82 kg. He was at a party and consumed 2 bottles of a certain alcoholic beverage, each equivalent to 1.9 standard drinks.

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He stopped drinking at 12:10 am and left the party at 2:40 am just when his blood alcohol content (BAC) reached zero.

The time needed for BAC to reach zero after the last drink consumed can be calculated using

$$\text{Number of hours for BAC to reach zero} = \frac{\text{BAC}}{0.015}$$

The formula to calculate BAC for a male is  $BAC_{\text{Male}} = \frac{10N - 7.5 H}{6.8 M}$  where N is the number of standard drinks consumed, H is the number of hours since he started drinking and M is his Mass in kg.

At what time did Patrick start to have his first drink at the party?

Give your answer to the nearest minute.

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**Question 33 (2 marks)**

A chicken coop contains 10 chickens. 6 of them are white and 4 are brown.

**2**

If three chickens are chosen at random, find the probability that more brown chickens than white are selected.

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**Question 34 (3 marks)**

The ratio of mango juice to water in a cup containing 210 mL of drink is 6: 1.

**3**

Karen added water to the cup. The ratio of mango juice to water in the cup became 9: 4.

How much water did Karen add to the cup?

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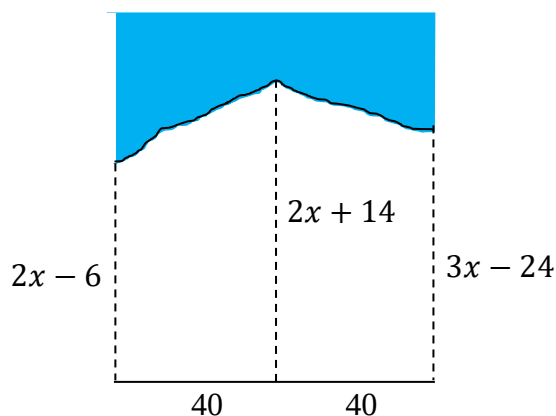
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### Question 35 (6 marks)

The diagram shows the land that Melanie bought near the ocean. All measurements are in metres.



- (a) Use two applications of the trapezoidal rule to show that the approximate area of the land is  $180x - 40$ .

3

[illegible]

- (b) If a quarter of the area of this land is  $50x - 150$ , find the area of this land.

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**END OF PAPER**

## Section II extra writing space

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## Section II extra writing space

If you use this space, clearly indicate which question you are answering.

[illegible]

## Section II extra writing space

[illegible]

## REFERENCE SHEET

### Measurement

#### Precision

Absolute error =  $\frac{1}{2} \times \text{precision}$

Upper bound = measurement + absolute error

Lower bound = measurement – absolute error

#### Length, area, surface area and volume

$$l = \frac{\theta}{360} \times 2\pi r$$

$$A = \frac{\theta}{360} \times \pi r^2$$

$$A = \frac{h}{2}(x + y)$$

$$A \approx \frac{h}{2}(d_f + d_l)$$

$$A = 2\pi r^2 + 2\pi rh$$

$$A = 4\pi r^2$$

$$V = \frac{1}{3}Ah$$

$$V = \frac{4}{3}\pi r^3$$

#### Trigonometry

$$A = \frac{1}{2}ab \sin C$$

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$c^2 = a^2 + b^2 - 2ab \cos C$$

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

### Financial Mathematics

$$FV = PV(1 + r)^n$$

#### Straight-line method of depreciation

$$S = V_0 - Dn$$

#### Declining-balance method of depreciation

$$S = V_0(1 - r)^n$$

### Statistical Analysis

$$z = \frac{x - \bar{x}}{s}$$

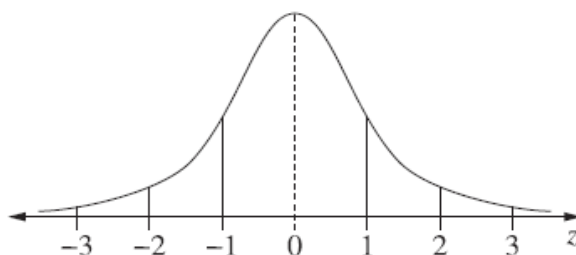
An outlier is a score

less than  $Q_1 - 1.5 \times IQR$

or

more than  $Q_3 + 1.5 \times IQR$

#### Normal distribution



- approximately 68% of scores have  $z$ -scores between  $-1$  and  $1$
- approximately 95% of scores have  $z$ -scores between  $-2$  and  $2$
- approximately 99.7% of scores have  $z$ -scores between  $-3$  and  $3$